

Team 39 Pitch for Eviden's Data driven sustainability

Background information on Eviden (for Team 39):

- Eviden is a global leading IT solutions provider. It was formed in 2023 as one of the largest carve-outs the industry has ever seen after Atos decided to split their company into 2 parts Eviden has a revenue of €5 Billion
- The company is a digital native scale-up and leader in data, cloud, advanced computing, and cybersecurity. Their deep industry expertise and proprietary technology accelerate digital transformation.
- Eviden's mission is to be your preferred digital, cloud, data, and security partner to help you build your digital future. They believe that the world's most pressing and complex challenges hold infinite possibilities and require solutions yet to be imagined. Eviden aims to transform possibility into reality and deliver what digital transformation promised.
- The company has a strong ethos. They dare to try and push each other to stay curious. They are united across the world and driven to achieve, with 57,000 employees worldwide, 50,000 digital certifications, 2,100 patented innovations, and €5b revenue in 2022.
- At Eviden, their mission is to create new opportunities and pave the way forward for their clients and society as a whole, both today and for generations to come.

Deliverables as per the Brief:

- Thoroughly investigate data-driven sustainability.
- Recommend cutting-edge best practices.
- Conduct comprehensive impact analyses of emerging environmentally-conscious IT technologies.
- Deliver a Proof of Concept (POC).

Part 1: Introduction

- In this section we will introduce the team participating in the pitch and possibly roles if relevant, also thank Eviden for the opportunity to present the project.
- We can also include any personal reasons why we are happy we got this project such as passion for the environment (climate change, net 0), passion for the company (Atos; 100,000 employees, €11 Billion in revenue, etc)
- We should also aim for this to be less than a minute long if possible due to more important sections

Part 2: Problem Statement

- Here we'll discuss the problem we're trying to solve and explain how we will deliver the main task of:
 - Selecting a use case that will demonstrate environmental sustainability value – not only the experience itself, but how it tackles the sustainability challenge.
 - Develop either a Proof of Concept (POC) or final solution, using an on prem or cloud platform and user interfaces (dashboard) to implement the chosen solution.
 - Showcase and demonstrate the POC or final solution.
- Can also explain why problem is important for Eviden and Tech industry as a whole
- Definition of Data-Driven Sustainability:
 - Use data and analytics to guide sustainable decision-making.
 - Involve collecting, analysing, and interpreting data for insights into environmental impact, resource usage, social responsibility, and economic viability.

Part 3: Solution

- This should be our explanation of how we want to solve this issue, we need to decide how much detail we want to give, but this will most likely be one of our longer sections we discuss.
- It's one of the things that will show how much we understand the project and set us aside from other teams.
- Example Use Cases the team has come up with:
 - Renewable Energy Integration:
 - Implement AI-driven systems to forecast energy demand and integrate renewable energy sources like solar panels and wind turbines, maximising the use of sustainable energy.
 - Can talk about economic and social impacts of renewable energy - creation of jobs, through factories, engineers etc -> Better for the economy overall, as more spending occurs -> QoL increases.

- Plastic Waste Reduction:
 - Create an AI-powered system that tracks and reduces plastic waste by monitoring product packaging, encouraging the use of biodegradable materials, and optimising recycling processes.
- Creating Software:
 - Develop software to consider multiple factors:
 - CO2 emissions produced vs. benefits of the decision.
 - Resource usage (minimization and optimization).
 - Long-term societal impact (benefits or drawbacks).
 - Job creation or impact on employment rates.
- Software's Role:
 - Predict the outcomes of decisions made to ensure they align with data-driven sustainability principles.

Suggested Approach:

1. Selecting a Use Case for Environmental Sustainability:
 - Choose a use case demonstrating environmental sustainability impact.
 - Example: Optimise data centre energy usage using AI algorithms for reduced carbon emissions.
2. Developing the Proof of Concept (POC) or Final Solution:
 - Platform Selection: Choose the appropriate cloud-based platform considering accessibility, cost, and carbon footprint.
 - Suggested platforms: Google Cloud Platform (GCP), Vertex AI for AI/ML, Talk about that later on - Azure.
 - Similar tech stacks available from other cloud providers (AWS, Azure).
 - User Interfaces: Develop user-friendly dashboards for interaction.
 - Mention relevant team experience
 - AI Integration: Integrate AI technologies for data analysis, pattern recognition, and predictive modelling.
3. Ensuring Security and Access Control:
 - Implement robust authentication and authorization systems for data management and security.
4. Company Communication and Project Planning:

- Ensure effective communication throughout the project, gathering feedback and aligning with business goals.
 - Develop a detailed project plan using tools like Gantt Charts and Git. Justify the chosen method and format.
 - Define specific, measurable objectives aligned with sustainability goals for progress tracking and accountability.
5. Monitoring and Evaluation:
- Continuously collect relevant data to assess the solution's impact on sustainability metrics.
 - Regularly evaluate the solution's performance against set objectives and make necessary adjustments.
6. Documentation and Knowledge Sharing:
- Maintain comprehensive documentation, including codebase, system architecture, and user manuals for knowledge retention.

Part 4: USPs and Team strengths

I feel as though we can merge our USPs and Team strengths sections as they interlink, we should highlight anyone with backgrounds in dealing with **Data Management/Handling, AI, Sustainability/Environmental issues, Optimisation, (Cyber) Security, Industry Experience**. I would personally opt for us to somewhat mention Andreas' website project as that involves working with databases and with a client, and Martin's industry experience and awareness of net 0 goals - how important we understand it is. Must also mention how our solution is superior.

Skills that we can mention:

- Andreas Sergiou - Website Project - communication with client, user-friendly interface, for website and for chat-bot - can show if they want proof.
- Aaron - Practical experience in artificial intelligence, having created a 3D video game in which the enemy AI learned and adapted to players' movements.
- Martin - Coffee Shop experience - Uses sustainable materials in their products and awareness of net 0 goals.
- Everyone - Can say that we all have web dev experience - Databases and Interfaces module in year 1 - HTML, CSS, JavaScript, PHP, SQL

How our product differs:

- Using AI that's trained on previous events that have happened, and knows the effect of the event - allows for a more accurate response and solution - more reliable.

Part 5: Working Style & Methodology

Our Methodology:

- **Agile development** - We are going to be using agile methodologies to ensure flexibility and responsiveness in case of any changing requirements throughout the project. This is an iterative approach, allowing us to adapt to any changes.
- **Version Control with Git** - use GitLabs for version control, allowing for easier collaboration, monitoring changes and ensuring the integrity of our codebase.
 - Make sure all team members know what changes have been made and how their changes affect the overall project.
 - We also use Git to assign team members issues, so they know what tasks they should be doing.
- **Regular Meetings and Documentation** - We schedule multiple team meetings every week, with the team admin - Andreas Sergiou - writing detailed meeting minutes documentations, tracking the progress of the project and ensuring all team members are aware of important decisions being made and what issues have been brought up.
- **Paired Programming** - Can mention this if they ask about how code review will work.
 - real-time collaboration, knowledge-sharing etc.

Choice of Programming languages and Cloud Platform:

- **Python for Machine Learning** - Taking team strengths into account and Charlotte Fardell's background in Python for machine learning, we plan to use Python's extensive libraries to develop the AI - analyse data etc.
- **Azure Cloud Services** - Taking into account Charlotte Fardell's Azure Cloud training and that it meets what we want to use it for, we should propose using Azure Cloud - ask what training we could do.
 - **Summary we can talk about**
 - Provides a range of AI and data analytics services - robust foundation for project - it's scalable and has integration capabilities.
 - **Benefits of Azure:**
 - **Integration Capabilities** - Azure integrates with various tools, allowing for smooth collaboration and data exchange between all components of our solution.
 - **AI and Machine Learning Services** - Azure has a suite of AI and machine learning services, allowing easier implementation of algorithms and predictive modelling.

- **Data Security** - Azure provides security measures - encryption, identity management etc.

- **Energy-Efficient Computing (sustainability point - we've thought of the environment with our proposed solution)** - Azure is committed to be carbon negative by 2030 - Aligns with Eviden's sustainability goals.

- **Data Center Locations** (Can relate this to our Use Case example) - Azure uses data centres in places powered by renewable energy -> lower carbon footprint etc

Part 6: Ethos and Values

This is our change to express how much we care about the task and environmental sustainability in the field of computer science. As per my own research, out of the two POCs, Charlotte Fardell is the Data Scientist and is also 'passionate about making a positive impact on the environment through my work and daily activities' - verbatim as per her LinkedIn about me. She is also 'currently undertaking Azure Cloud training', I believe we should heavily consider using Azure Cloud for this reason, but of course if another service suits us more we should go for that instead. Also from what I've seen, they use python for machine learning practices.

Part 7: Next Steps and Timeline

Questions we could ask Eviden:

- How will they support us throughout this project?
- How often do they expect us to be in contact with them - to give them updates.
- If there are any features in particular that they want to see in the project.
 - Features they believe to be critical in the success of the project?
- Ask about the Azure Cloud Training our other point of contact - Charlotte is doing - if they recommend us to do it as well - we get a certificate - become more certified?
- Are there existing systems or platforms within Eviden that our solution needs to be integrated with?
- Will there be an opportunity for end-users or stakeholders within Eviden to provide feedback? - Can cater the final product to their needs.
- What motivates the team to work on this project for Eviden?

Questions we could be asked:

- Q1) Why would we use Azure?
- A1) We plan to use Azure as it provides a range of AI and data analytics services - robust foundation for the project. Some benefits we've identified is its Integration

Capabilities - Azure integrates with various tools, allowing for smooth collaboration and data exchange between all components of our solution.

- **AI and Machine Learning Services** - Azure has a suite of AI and machine learning services, allowing easier implementation of algorithms and predictive modelling.

- **Data Security** - Azure provides security measures - encryption, identity management etc.

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- What motivates the team to work on this project for Eviden?

- Passion for sustainability

- Admiration for EVIDEN → company mission of leveraging tech and data to solve problems.

- Relevance of the project in today's society.

- Develop on our use case that we mentioned?

- We plan to first identify a range of potential use cases.

- We will then analyse each use case, and pick one that is most sustainably relevant and aligned with our values.

- Through historical data, industry benchmarks and relevant metrics.

- What is the suggested approach for selecting a cloud platform?

- We considered multiple key factors:

- Accessibility

- Cost-effectiveness

- Platform's carbon footprint → 100% renewable energy by 2025.

- Integration capabilities

- Azure offers a scalable and cost-efficient infrastructure.

- What methodology is Team 39 planning to use for project development, and why is it chosen? - We are planning to use an Agile development methodology, for several reasons:

- Flexibility and responsiveness → adaptiveness to changing requirements + circumstances

- Iterative approach → Breaking project down into smaller, easily manageable pieces.

- What version control system is mentioned, and how is it intended to be used by the team? - Git enables us to work collaboratively on the project together, through remote collaboration, we are not bound to having to meet every time a important decision needs to be made.

- How will AI technologies be integrated into the solution, and what is their role? - We will use AI to analyse and process data. Use examples such as positives and negative repercussions of placing a sustainable power plant in a given area.

- What steps are being taken to ensure security and data protection in the project? - We will implement security through choosing a secure platform to store data, and rigorous testing of the solution.

- How will the solution's impact on sustainability metrics be monitored and evaluated? - We can measure the number of kilograms of carbon emitted, and display it in the user dashboard of the solution.

- What is the significance of maintaining comprehensive documentation in this project? - It is of great significance to maintain comprehensive documentation such that EVIDEN know what our plans are every step of the way to minimise ambiguity and disagreements between our parties.

- How does the team express its commitment to sustainability in the field of computer science? - Team members are involved and familiar with sustainable practices. Our awareness of environmental impacts guides our approaches to technology solutions.

Able to engage in a more personal talk with the team at Eviden. Could also provide a timeline for key milestones.

Part 8: Conclusion and Call to Action:

- A small speech type thing someone can say:

'In conclusion, our team is not only dedicated, but deeply passionate about driving positive changes through sustainable tech solutions. We firmly believe that our proposed approach, rooted in cutting-edge technologies and softwares and shared commitment to positive environmental change, will exceed your expectations. Our team is ready and eager to get started on this project with you, working hand-in-hand and creating a revolutionary end product. We thank you for considering us and giving us the time to express our interest in your project.'

Video Script

*[Opening Shot - Andreas, Aaron and Mateusz are stood in front of the Jubilee Conference Building - facing the conference building so the lake is in the background or visa versa]
[All standing in a semicircle, relaxed but in a professional manner - Camera pans in smoothly as the video begins]*

[Background Music - Upbeat and Inspirational]

Part 1 - Introduction:

Andreas - Hello and welcome, Eviden team! We are thrilled to have the opportunity to present to you our vision for a sustainable future. My name is Andreas Sergiou, the Team Admin of the Group, and I am joined by my teammates, Aaron and Mateusz. We are passionate about this project for various reasons, from our commitment to the environment to our admiration for Eviden's mission and values.

[Camera pans to Aaron and Mateusz - nodding in agreement]

Part 2 - Problem Statement:

[Shot - Footage of environmental challenges/stock footage of sustainability stuff - Can be inside around a table/screen to show the slides]

Aaron - Now let's delve into the problem. We're here to address the challenges of data-driven sustainability. Our goal is to recommend the best practices, analyse emerging environmentally-conscious IT technologies and ultimately deliver a software that predicts the outcome of 'decisions' and seeing if they are done in a 'data-driven', sustainable manner and what their impacts will be.

[Shot: mission statement]

Mateusz - Eviden's mission and our project's relevance couldn't be more relevant in today's society. Sustainability is crucial for Eviden and the tech industry as a whole. Data-driven sustainability involves using data and analytics to guide sustainable decision-making.

Part 3 - Solution:

[Shot: Visual representation of the example use cases]

Andreas - We've come up with several use cases, like renewable energy integration, plastic waste reduction, and creating software that considers various factors for decision-making. Our solution will predict outcomes and correspond with sustainability principles, rooted in data.

[Shot: Visuals of Azure tech stack]

Aaron - We're suggesting an approach that selects a use case that demonstrates significant environmental sustainability impact, such as optimising data centre energy usage using AI algorithms to reduce carbon emissions.

Mateusz - We will carefully choose a cloud-based platform on accessibility, cost, and carbon footprint considerations. Intuitive dashboards will be developed to ensure user-friendly interaction. We are planning to use the Azure tech stack, for its integration with various tools and AI and machine learning services, allowing easier implementation of algorithms and predictive modelling.

Andreas - When applicable, we will integrate AI technologies for data analysis, pattern recognition, and predictive modelling.

Aaron - We plan to implement a robust authentication and authorization system, ensuring data encryption occurs to protect sensitive data.

Mateusz - We will continuously collect relevant data to assess the solution's impact on sustainability metrics. Regular evaluation against set objectives, along with comprehensive documentation, will ensure deadlines are met and expectations are exceeded.

Part 4 - Unique Selling points and Team Strengths:

[Shot: Visual of the team members at work]

Mateusz - Our team brings a diverse set of strengths to the table. Andreas has experience in web development and client communication, Aaron has practical AI experience and I have back-end experience, using PHP and SQL querying.

[Shot: One more shot of the team members at work at a different angle]

Part 5 - Working Style and Methodology:

[Shot: needs some sort of filler here - can add something from the slides]

Aaron - We've decided to use Agile development for this project, as it allows for flexibility and responsiveness in case of any changing requirements throughout the project. Git is also being used for version control, allowing for easier collaboration, monitoring changes and ensuring the integrity of our codebase.

Part 6 - Ethos and Values:

[Shot: Visual reflecting sustainability efforts - cleaning up the ocean/ planting trees etc]

Andreas - We believe technology should empower positive change and embrace adaptability, collaboration, and continuous learning. Beyond technical expertise, we prioritise effective communication, transparent collaboration, and empathetic understanding. Our mission extends beyond this project; we're dedicated to a greener, more sustainable future

Part 7 - Conclusion and Call to Action:

[Closing Shot - Andreas, Aaron and Mateusz are stood in front of the Jubilee Conference Building - facing the conference building so the lake is in the background or visa versa - same location as the opening shot]

Mateusz - In conclusion, our team is fully dedicated to creating a positive environmental change. We believe our approach coupled with Eviden's values, will exceed your expectations. We are ready and eager to get started. Thank you for considering us!

[Background music dies down and the screen fades to solid white - the page can have the team 39 logo]

Other Ideas:

- One shot can be a walking shot - walking along the lake

